Market-Implied Inflation Forecasts from Kalshi CPI Prediction Markets

Accuracy, Calibration, and Comparison Against a Professional Benchmark

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Abstract

We evaluate whether Kalshi's Consumer Price Index year-over-year (CPI YoY) prediction markets produce accurate, well-calibrated inflation forecasts, and how they compare to a professional benchmark. Using 44 monthly releases from November 2022 to June 2026, we convert threshold-contract prices into implied probability distributions, then score them against realized CPI at six pre-release horizons. The market is highly informative (mean Brier score approximately 0.07, well below the 0.25 coin-flip reference), grows both more accurate and more confident as each release approaches, and calls the direction of inflation correctly about 97 percent of the time. Naive baselines score near 60 percent, indicating the market's directional skill is genuine rather than a byproduct of inflation's persistence. On point-forecast accuracy the market is competitive with, and slightly ahead of, the Federal Reserve Bank of Cleveland's inflation nowcast. We interpret these findings as strong evidence over a single inflationary regime rather than definitive proof.

1 Introduction

Online prediction markets such as Kalshi convert public expectations about a future event into a single observable price. Kalshi lists contracts on the Bureau of Labor Statistics' monthly Consumer Price Index year-over-year release (CPI YoY) well before each official announcement. Because these contracts resolve to \$1 or \$0 based on the realized outcome, their pre-release prices can be interpreted as the market's implied probability that inflation will exceed a given threshold.

This project asks one central question: how accurate and well-calibrated are these market-implied forecasts, relative to a real macroeconomic outcome and relative to an established professional benchmark? We focus on CPI YoY specifically because it is one of the most important and closely watched macroeconomic releases; because Kalshi lists dense ladders of threshold contracts for each release, giving a rich cross-section of implied probabilities; and because the monthly cadence provides ample history to form meaningful evaluations over time.

The paper is organized as follows. Section 2 describes our data sources. Section 3 explains the pipeline that converts Kalshi's raw market prices into standardized forecast distributions. Section 4 describes our scoring methodology. Section 5 presents results, Section 6 discusses limitations, and Section 7 concludes.

2 Data

2.1 Kalshi market data

All primary data was pulled through Kalshi's public REST API (KXCPIYOY series), which does not require authentication for market-data reads. The series comprises 44 monthly CPI YoY releases spanning November 2022 through June 2026. Each release is represented as a threshold ladder of "Above X%" contracts. Each contract's price sits between $0 and $1 and is interpreted directly as the market-implied probability that realized CPI YoY exceeds that threshold. Prices were pulled as daily candlesticks, capturing open, high, low, and close for both traded price and bid/ask quotes.

A notable API characteristic encountered during data collection was that Kalshi serves recent data (roughly the last three months) through one set of endpoints and older data through a structurally distinct set of historical endpoints, with different field-name conventions, despite both covering identical underlying data. The pull pipeline was therefore built to detect and reconcile both schemas; the fallback procedure is described in Section 3.1. Handling this split was essential: a naive pull that used only the live endpoints returned just the two most recent events, whereas the full history required the historical fallback.

2.2 Independent validation series

As a consistency check, we additionally pulled Kalshi's KXECONSTATCPIYOY series, which represents the same underlying CPI outcome as a set of mutually exclusive "exactly X%" bins rather than a cumulative threshold ladder. This series has a shorter history, from December 2025 to June 2026, but provides an independently priced cross-check on our primary series' implied probabilities.

2.3 Benchmark data

Realized CPI YoY values were pulled directly from the BLS public API (series CUUR0000SA0, all items, not seasonally adjusted), computing the year-over-year percentage change with the same half-up rounding convention the BLS applies to its published headline figure. Of the 44 releases in scope, 43 had been published as of data collection; the only missing value was October 2025, which was never published owing to the government shutdown.

As an external professional benchmark, we used the Federal Reserve Bank of Cleveland's Inflation Nowcasting series, sampled at the same horizons as the Kalshi-derived forecasts for a matched comparison. The nowcast is published daily; we used its real-time vintages, so that each value reflects only information available before the corresponding release.

3 Forecast Construction Methodology

The pipeline converts raw market prices into a standardized, scorable historical forecast panel in five main steps.

3.1 Data ingestion

For each of the 44 events, we retrieved the full set of threshold contracts and their complete daily price history. Because Kalshi serves recent and older events through different endpoint families, the pipeline first queried the live endpoints for each event and, when those returned no market data (which occurs for all but the most recent few events), automatically fell back to the corresponding historical endpoints. Each event was tagged with the source it was ultimately served from, so that coverage could be audited. Where a contract's price fields used the historical schema's field names, these were then mapped to the equivalent live-schema fields prior to any downstream processing, ensuring a single consistent representation regardless of data source.

3.2 Flattening and contract parsing

The nested per-event JSON records were flattened into a single long-format table with one row per (event, contract, day). On days with no executed trade (approximately 57 percent of contract-days in the dataset), the last-traded-price field is undefined; on these days we substitute the midpoint of the bid and ask quotes as the probability proxy, since standing quotes persist even without a completed trade. Each contract's threshold value was parsed primarily from Kalshi's own contract metadata (`floor_strike` / `cap_strike` fields), with ticker-string parsing used only as a documented fallback when metadata was unavailable (roughly 10 percent of contracts).

3.3 Fixed-horizon snapshots

Because different releases had markedly different amounts of trading history, we standardized every release to six fixed evaluation horizons: 21, 14, 7, 3, and 1 calendar day before release, plus a final pre-release snapshot. At each horizon we selected the most recent eligible price observation at or before the target date, recording both the resulting probability and the staleness of that observation (days between the target horizon and the observation actually used). Observations older than 14 days relative to their target horizon were kept and flagged as potentially unreliable rather than discarded, to preserve transparency about data quality.

3.4 Survival curve construction and monotonicity correction

For each event and horizon, the set of threshold-contract prices forms a survival curve $S(t) = P(\text{CPI} > t)$, which should be weakly decreasing in t. Raw market prices, however, are subject to noise and asynchronous updates across contracts, which can produce local violations of this property. We identified and corrected such violations using isotonic regression, which fits the closest weakly-decreasing curve to the observed prices. Of 261 event-horizon curves evaluated, 149 (57 percent) contained at least one monotonicity violation, comprising 318 individual violations across roughly 3,300 adjacent threshold-pair comparisons, a per-comparison violation rate of approximately 10 percent. We note that although violations were common, the individual corrections were typically small; isotonic regression redistributes the adjustment optimally rather than merely capping values downward, avoiding the directional bias that a simpler running-minimum rule would introduce.

3.5 Differencing and summary statistics

The monotonicity-corrected survival curve was differenced across adjacent thresholds to obtain a proper probability mass function over one-decimal CPI outcomes, with explicit open lower and upper tails. From this distribution we computed the median (via linear interpolation across the 50 percent crossing), the modal outcome bin, tail probabilities, an implied mean, and, derived from the same curves, the distribution's standard deviation and interval width. Because the outermost bins are open-ended, the implied mean requires a documented tail assumption: we treat each open tail as centered one half bin-width beyond the outermost observed threshold. This is a simplification for summary purposes, not a modeled tail distribution, which is why we lead with the median (which requires no such assumption) in the directional analysis.

3.6 Independent cross-series validation

As a validation step prior to scoring, we compared the primary series' differenced bin probabilities against KXECONSTATCPIYOY's directly quoted bin probabilities, matched by release month and evaluated across all six horizons. Across 624 bin-level comparisons spanning seven overlapping months, the two independently priced series exhibited an overall correlation of 0.66, providing external evidence that the construction methodology produces probabilities consistent with an independently priced source, rather than merely internally self-consistent output.

4 Scoring Methodology

Following guidance to prioritize interpretable, low-complexity accuracy measures over more elaborate statistical machinery, we scored the corrected forecast panel using two primary measures, plus a benchmark comparison.

4.1 Brier score

For each threshold contract we define a binary outcome (1 if realized CPI exceeded the threshold, 0 otherwise) and compute the squared error between the corrected probability and this outcome. The Brier score is the mean of this squared error across all threshold contracts for a given event and horizon, averaged across all scored releases. A Brier score of 0 indicates perfect, fully confident calibration; a score of 0.25 corresponds to an uninformative 50/50 forecast and serves as our reference baseline. The threshold outcome is coded on an at-or-above convention (a strike exactly equal to the realized value counts as exceeded); this affects only the rare release landing exactly on a listed threshold.

4.2 Directional accuracy

For each release and horizon we derive a directional call, whether the market's median implied forecast anticipated an increase or decrease in CPI YoY relative to the prior release, and compare this call against the realized direction. We use the median rather than the mean as the point forecast because it is read directly from the survival curve at the 50 percent crossing and requires no assumption about the open-ended tails. We report both an unweighted hit rate (each call counted equally) and a confidence-weighted hit rate, in which each call is weighted by the magnitude of the predicted move, so that high-conviction calls carry proportionally more weight.

4.3 Benchmark comparison

We compared the market's directional accuracy and point-forecast error against four alternatives, sampled at the same months and horizons: the Cleveland Fed inflation nowcast (a professional statistical model); a naive momentum baseline (predicting the same direction as the most recent realized change); a naive majority-class baseline (always predicting the historically more common direction); and a persistence, or random-walk, baseline (forecasting that next month's CPI equals the current month's) for the point-accuracy comparison. To ensure a fair test, all methods were scored on the common set of releases for which every method had a forecast and the realized direction was not flat; this yields between 37 and 39 releases per horizon, fewer at the longest horizons where a benchmark may not extend far enough back.

5 Results

We scored 44 monthly releases (43 with a published realized value; October 2025 was excluded as unpublished). We organize the findings around the three questions posed in the introduction: is the market accurate, does it sharpen over time, and does it beat established benchmarks?

5.1 The market is accurate, and sharpens as the release approaches

Table 1 reports the mean Brier score at each horizon. Every value lies far below the 0.25 coin-flip reference, indicating the market's probabilistic forecasts are genuinely informative. The score also improves steadily as the release approaches, from approximately 0.084 twenty-one days out to 0.064 three days out, and then plateaus in the final days. This pattern is the signature of a forecaster incorporating information over time and settling once most uncertainty is resolved.

Table 1: Mean Brier score by horizon (lower is more accurate; 0.25 = coin flip).

Horizon	Mean Brier score	Interpretation
21 days	0.084	least accurate horizon
14 days	0.071	
7 days	0.070	
3 days	0.064	most accurate horizon
1 day	0.065	
final	0.065	plateau

5.2 The market also grows more confident, in step with its accuracy

Beyond point accuracy, the market's implied distribution tightens as the release nears. Table 2 reports the mean 80 percent interval width and standard deviation of the implied distribution by horizon; both decline monotonically, from a standard deviation of about 0.30 percentage points twenty-one days out to about 0.19 one day out. Crucially, this sharpening of confidence tracks the improvement in accuracy from Section 5.1: the market becomes both more confident and more correct at the same time. A forecaster that narrowed its distribution without a corresponding gain in accuracy would be exhibiting overconfidence; the co-movement of the two is instead consistent with genuine information incorporation.

Table 2: Spread of the market's implied distribution by horizon (narrower = more confident).

Horizon	Mean interval width (80%)	Mean std. deviation
21 days	0.41	0.30
14 days	0.34	0.25
7 days	0.33	0.21
3 days	0.32	0.20
1 day	0.32	0.19

5.3 Directional skill is real, and competitive with a professional model

The market called the direction of inflation, higher or lower than the prior release, correctly about 97 percent of the time, with a confidence-weighted hit rate near 99 percent, pooled across horizons.

A 97 percent hit rate warrants scrutiny, because inflation is persistent and its direction is therefore often easy to guess. To separate genuine skill from persistence, we benchmarked against naive rules that use no market information. As Table 3 shows, the momentum and majority baselines score only about 64 and 59 percent respectively. The market's roughly 37-point margin over the best naive rule cannot be explained by inflation's trend alone, and it holds at every horizon, including twenty-one days before release.

Against the Cleveland Fed nowcast, a professional statistical model, the market edges ahead on direction (about 97 versus 92 percent at the final horizon) and is closer on point accuracy: its mean absolute error one day before release is 0.075 percentage points, versus the nowcast's 0.122 and a naive random walk's 0.333. We characterize the market as competitive with, and slightly ahead of, the professional benchmark, rather than decisively superior, given the narrowness of the directional margin on a sample of this size.

Table 3: Directional accuracy and point-forecast error, market versus benchmarks.

Method	Direction (final)	Direction (21 d)	MAE (1 d, pts)
Kalshi market	**97%**	**95%**	**0.075**
Cleveland Fed nowcast	92%	89%	0.122
Momentum (naive)	64%	62%	–
Majority (naive)	59%	57%	–
Persistence (random walk)	–	–	0.333

The market's directional calls were also most confident precisely when they were correct. Table 4 reports the unweighted and confidence-weighted hit rates by horizon; weighting each call by the magnitude of the predicted move raises the hit rate to essentially 100 percent at every horizon from fourteen days out, indicating that the market placed its largest directional bets on the releases it called correctly.

Table 4: Directional hit rate, unweighted versus confidence-weighted, by horizon.

Horizon	Unweighted hit rate	Confidence-weighted
21 days	95%	99.5%
14 days	97%	99.9%
7 days	97%	100%
3 days	100%	100%
1 day	97%	100%
final	97%	100%

Table 5 extends the point-forecast comparison across all six horizons, reporting both mean absolute error (MAE) and root mean squared error (RMSE), the latter penalizing large misses more heavily. The market's error declines as the release approaches, mirroring the Brier and confidence results, and it is roughly three to four times smaller than the persistence (random-walk) benchmark at every horizon. The market's advantage holds under RMSE as well as MAE, confirming the result is not driven by a handful of outlier releases. The Cleveland Fed nowcast's point-forecast error was evaluated at the one-day horizon, where the market's MAE of 0.075 percentage points is lower than the nowcast's 0.122; a full per-horizon nowcast comparison is left to future work.

Table 5: Point-forecast error by horizon: market versus persistence baseline.

Horizon	Market MAE	Market RMSE	Persistence MAE	Persistence RMSE
21 days	0.097	0.143	0.293	0.399
14 days	0.083	0.115	0.295	0.399
7 days	0.082	0.108	0.295	0.399
3 days	0.069	0.094	0.302	0.404
1 day	0.072	0.096	0.302	0.404
final	0.072	0.096	0.302	0.404

Taken together, the three results answer the central question in the affirmative: the market is accurate and well-calibrated, it sharpens as each release approaches on two independent measures, and its skill exceeds naive baselines by a wide margin while remaining competitive with a professional forecasting model.

6 Limitations

Several limitations bound the strength of these conclusions and motivate the framing of our results as strong evidence rather than proof.

Sample size and dependence. Our evidence rests on 44 monthly releases within a single, largely disinflationary regime. Year-over-year prints overlap by eleven months and inflation is highly persistent, so the observations are not independent. We therefore treat reported percentages as indicative rather than precise, and any formal inference would need to account for this autocorrelation (for example, via block or event-clustered standard errors).

Regime specificity. Because the sample covers one inflationary episode, we cannot claim the result generalizes to higher-volatility or structurally different regimes. Whether the market's edge persists in other conditions is an open question.

Market microstructure. Prices are treated as implied probabilities, but fees, bid-ask spreads, and risk premia sit between a contract price and a true probability. In addition, a substantial share of contract-days lack an executed trade and rely on quote midpoints, and thin, older markets exhibit frequent one-sided quotes; our monotonicity analysis found violations on 57 percent of curves, although the individual corrections were typically small.

Tail assumption. The implied mean depends on an assumption about the open-ended outer bins. We mitigate this by leading with the median and interval coverage, which do not require it, but the mean-based figures inherit this simplification.

Benchmark significance. While the market's advantage over naive baselines is large and consistent, its margin over the Cleveland Fed nowcast is narrow. We did not conduct a formal test of forecast-accuracy equality (for example, a Diebold-Mariano test) for that specific comparison, and on a sample of this size such a margin may not attain statistical significance at every horizon. We therefore state the professional-benchmark result conservatively.

7 Conclusion

We asked whether Kalshi's CPI year-over-year prediction markets produce accurate, well-calibrated inflation forecasts, and how they compare to an established professional benchmark. Across 44 monthly releases, the market emerges as a genuinely informative forecaster: its Brier scores sit far below the coin-flip reference and improve as each release approaches, its implied distribution tightens in step with that improving accuracy, and it identifies the direction of inflation correctly in roughly 97 percent of releases, far above naive baselines and competitive with, and slightly ahead of, the Federal Reserve Bank of Cleveland's inflation nowcast.

The central contribution is not merely that the market forecasts well, but that its skill survives the natural skeptical challenge. Because naive trend-following rules score near 60 percent while the market scores 97 percent, the market's directional accuracy reflects real information aggregation rather than the ease of extrapolating a persistent series. That a venue of participants trading modest sums matches a professional econometric model is a meaningful, if regime-bounded, statement about the forecasting value of prediction markets.

Natural extensions follow directly from the limitations. A larger, multi-regime sample and additional macro markets would test robustness; formal forecast-comparison tests would quantify the significance of the market's edge over the professional benchmark; the addition of professional-forecaster survey data would provide a second expert comparison; and a deeper calibration analysis, extending beyond central accuracy to the honesty of the full predictive distribution, would round out the evaluation. We leave these to future work, and offer the present study as strong initial evidence that a real-money prediction market can forecast inflation about as well as the experts.